MACHINE LEARNING PROJECT

Predicting Restaurant Ratings with ML

Random Forest vs Gradient Boosting on 9,551 restaurants: which model predicts customer ratings best?

95.2%

R² score of the best model

Gradient Boosting Regressor · MAE 0.216 on a 5-point scale

9,551	7	2
restaurants	features	models

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THE CHALLENGE

Can we predict a rating?

Restaurant ratings drive footfall, reviews and revenue. This project builds a regression model that estimates a restaurant's aggregate rating (0 to 5) from its business attributes: cost, price level, popularity, services and location.

9,551
Restaurants analysed

21
Columns in raw data

7
Model features used

80 / 20
Train / test split

Target variable

aggregate_rating

Continuous value from 0.0 to 5.0 a regression problem

METHODOLOGY

The ML pipeline

1 — Load & Clean

Loaded cleaned_data.csv and inspected columns and shape.

2 — Select Features

Kept 7 predictive columns; dropped city and cuisines.

3 — Split Data

80% train (7,640) and 20% test (1,911), random_state = 42.

4 — Encode

Mapped Yes/No columns to 1/0 so every input is numeric.

5 — Train Models

Fit Random Forest and Gradient Boosting regressors.

6 — Evaluate

Compared MAE, MSE, RMSE and R^2 on unseen test data.

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FEATURE ENGINEERING

What the model sees

average_cost_for_two
Typical spend for two people
Numeric

price_range
Price tier from 1 to 4
Ordinal

votes
Number of customer votes
Numeric

has_table_booking
Table booking offered
Yes/No 1/0

has_online_delivery
Online delivery offered
Yes/No 1/0

is_delivering_now
Currently delivering
Yes/No 1/0

country_code
Country of the restaurant
Numeric code

Dropped: city, cuisines (and all ID / address columns)
Every feature is numeric, ready for tree-based models.

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ALGORITHMS

Two ensemble models

R

Random Forest

Bagging ensemble

Many decision trees trained in parallel; predictions are averaged to reduce variance.

HYPERPARAMETERS

n_estimators	100
random_state	42
n_jobs	-1

G

Gradient Boosting

Boosting ensemble

Trees built one after another, each correcting the errors of the previous ones.

HYPERPARAMETERS

n_estimators	100
max_depth	5
random_state	42

Both models trained on the same 7,640 rows and tested on the same 1,911 rows

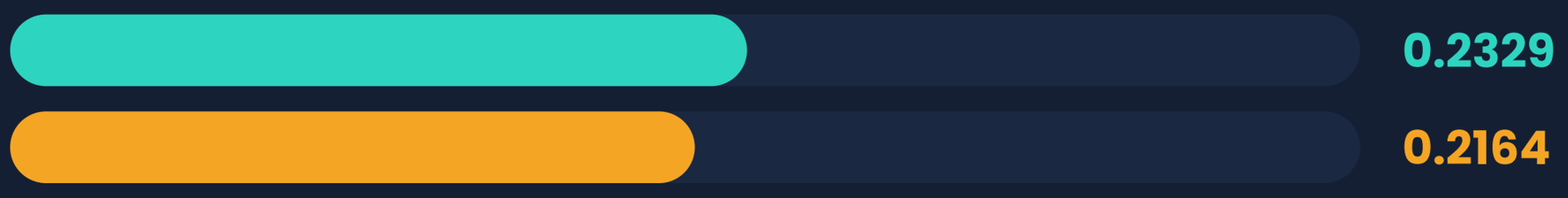
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RESULTS

Model showdown

Random Forest Gradient Boosting lower is better (errors)

MAE



MSE



RMSE



R² Score



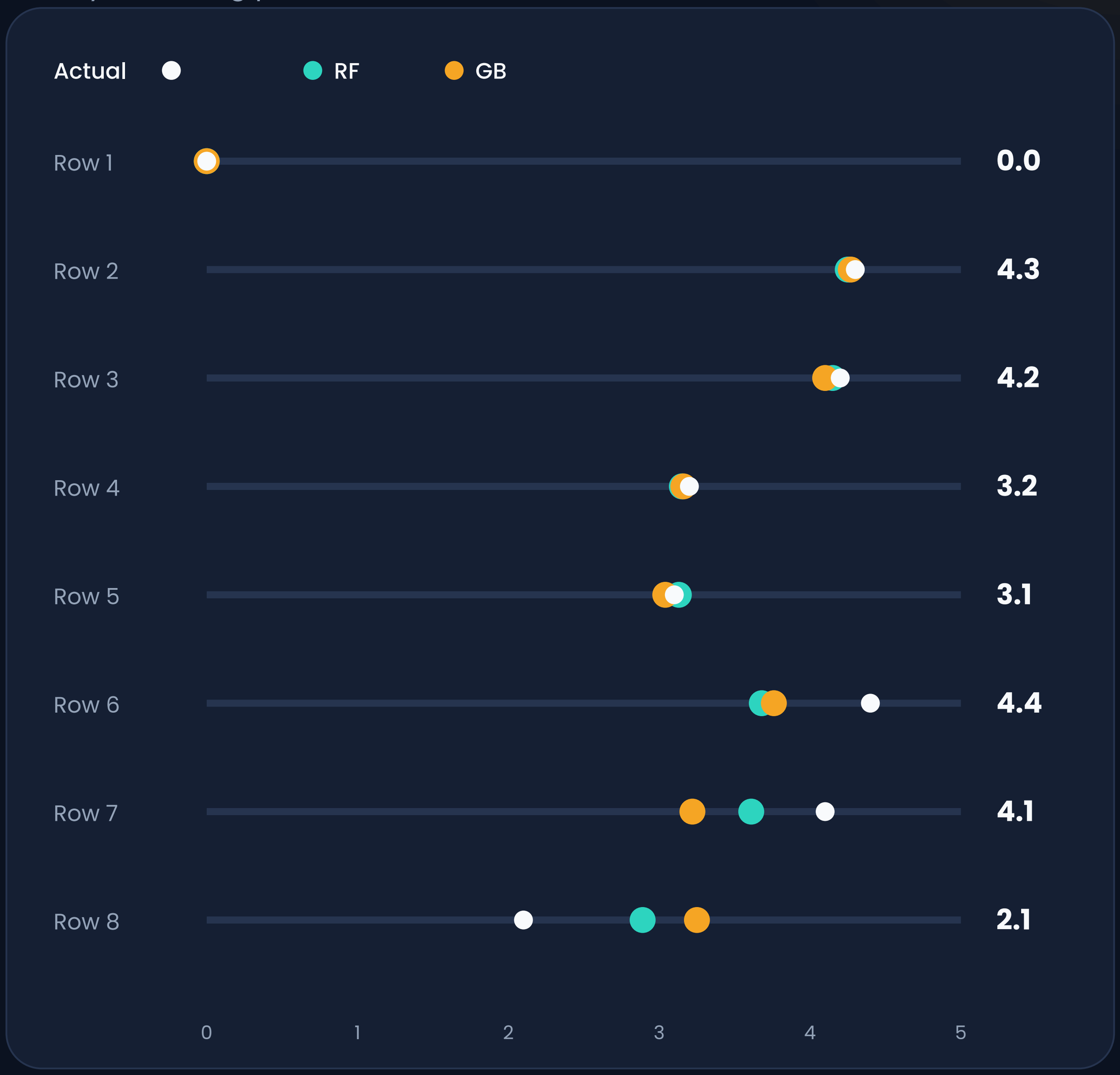
Gradient Boosting wins on every metric

MAE down 7.1% · RMSE down 8.6% · R² up 0.95 pts

REAL PREDICTIONS

Actual vs predicted

Sample rows from the 1,911-row test set. Both models track the true rating closely, including perfect hits on unrated restaurants (0.0).



Rows 1-5 hit closely. Rows 6-8 show the harder cases, where the model under- or over-shoots.

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CONCLUSION

Key takeaways

Best model

Gradient Boosting: R^2 95.18%, MAE 0.2164, RMSE 0.3313.

Strong accuracy

Average error is about 0.22 rating points on a 0 to 5 scale.

Simple inputs

Only 7 business attributes were needed; no text or review data.

NEXT STEPS

- Cross-validation and GridSearch / RandomSearch tuning
- Try XGBoost / LightGBM for further gains
- Add city and cuisines features with encoding
- Review feature importance to confirm no overreliance on votes

Tech stack

Python · Pandas · NumPy · scikit-learn · Matplotlib · Seaborn · Jupyter